Brief Statement about Dr Animesh Sinha

I have made contributions to research in forest genetics and tree improvement, particularly on Teak, Neem, Mahua, Kusum, bamboo, and RET species across eastern India. I have handled several projects (funded by ICFRE, CAMPA, DBT, and State Forest Departments) as Principal Investigator or Co-PI, focusing on genetic improvement, clonal propagation, and conservation of economically important and threatened species.

Large sets of candidate plus trees of Neem, Mahua, and Kusum were identified across several agro-climatic zones, variation in oil content and other key traits was quantified, and multivariate approaches such as exploratory factor analysis and CRITIC-based scoring were applied to develop statistically robust selection frameworks.

Protocols for *in vitro* conservation through slow-growth culture were developed for seven RET species, and tissue-culture protocols for several Bamboo species were standardized, thereby strengthening ex situ conservation and mass-multiplication capabilities. Tissue-culture-raised Teak clones have been developed and supplied in large numbers to farmers and the State Forest Department for establishment of field trials and production plantations.

The establishment and management of vegetative multiplication gardens, clonal trials, progeny trials, bambusetum, mangrove germplasm bank, and agroforestry research sites have direct extension relevance, as these serve as demonstration and resource sites for field functionaries and users. The technology on "Clonal propagation of mature Kusum (*Schleichera oleosa*) trees through cleft grafting" was recognized by ICFRE, Dehradun, for extension among farmers and upscaling for stakeholder popularization. A cost-effective *in vitro* propagation and conservation protocol for *Embelia ribes* Burm. f. was also developed.

I have played a substantive role in training and human resource development in forestry and allied disciplines. Seven Ph.D. scholars have been supervised to completion, and dissertation work of several M.Sc. and B.Sc. students from forestry, biotechnology, botany, and allied subjects has been guided through short-term (one to three months) project-based training, particularly in plant tissue culture and related techniques.

Curriculum vitae

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EDUCATIONAL DETAILS

Examination Passed	Board/Univ.	Year	Subjects
B.Sc. (Ag.) Hons.	Visva-Bharati, Santiniketan, West Bengal	1992	Agriculture Sciences
M. Sc.(Ag)	GBPant Univ. Agri & Tech., Pantnagar, Uttar Pradesh	1995	Plant Breeding & Genetics
NET(ICAR)	ASRB, New Delhi	1995	Plant Breeding
Ph.D.	Banaras Hindu Univ., Varanasi, Uttar Pradesh	2003	Genetics & Plant Breeding

SPECIALIZATION

- Forest Genetics Tissue Culture/ Tree Breeding/ Mangrove conservation

PUBLICATIONS

- **Journal Article (International and National):** 54
- **Book / Chapter/Bulletin:** 14

CITATION METRICS

- **Citation: 237**
- ***h*-index: 8**
- ***i10*-index: 6**

List of Publications

Paper published in Journal:

1. Malakar, A., Dutta, S., Shukla, N., Dasgupta, S., Singh, R. P., Dwivedi, S., ... & **Sinha, A.** (2025). Genetic variability and ecosystem services of Neem (Azadirachta indica A. Juss): A two-decadal systematic study for harnessing global sustainable goals. Trees, Forests and People, 22(2025), 101014. https://doi.org/10.1016/j.tfp.2025.101014
2. Malakar, A., & **Sinha, A.** (2025). Plus Tree selection for improvement of Neem (Azadirachta indica A. Juss): A heuristic approach through Exploratory Factor and CRITIC analysis. Forest Ecology and Management, 597(2025), 123146. https://doi.org/10.1016/j.foreco.2025.123146
3. Nayak, H., **Sinha, A.,** & Bhol, N. (2025). Genetic Diversity Assessment of Teak (*Tectona grandis* Linn. F.) in a Clonal Seed Orchard. *Journal of Sustainable Forestry*, 44(6), 319–333. https://doi.org/10.1080/10549811.2025.2516823
4. Malakar, A., Pradhan, M., Garai, S., & **Sinha, A.** (2025). Azadirachta indica A. Juss evinced robust resilience to changing climate under shared socioeconomic pathway scenarios in Eastern India. *Climatic Change*, *178*(4), 66. https://doi.org/10.1007/s10584-025-03910-x
5. **Sinha, A.**, Malakar, A., & Gorain, S. (2025). Efficacy of Edible Bamboo (Dendrocalamus asper) Based Agroforestry to Aid Socio-economic Status in Red and Lateritic Belt of Eastern India. *Indian Journal of Economics and Development, 21*(1), 69-76. https://doi.org/10.35716/IJED-23576
6. Kumari, R., Rathna, V., Kumari, G., **Sinha, A.**, & Mishra, Y. (2025). Molecular identification and elimination of endophytic contamination using antibiotics from in vitro culture of *Vitex peduncularis* Wall. Ex Schauer. *Plant Cell, Tissue and Organ Culture (PCTOC), 160*(2), 40. https://doi.org/10.1007/s11240-025-02975-x
7. Malakar, A., & **Sinha, A.** (2025). Genetic Variability Assessment of Azadirachta indica A. Juss in Eastern India: Implications for Tree Improvement. *Environmental and Earth Sciences Proceedings, 31*(1), 13. https://doi.org/10.3390/eesp2024031013
8. Malakar, A., Kumar, R., Mishra, Y., & **Sinha, A.** (2024). Evaluation of Seed Germination of Chironji (*Buchanania cochinchinensis* (Lour.) M. R. Almedia) under Nursery Conditions. *International Journal of Economic Plants, 11*(Nov, 4), 531–538. https://doi.org/10.23910/2/2024.5703b
9. **Sinha, A.,** Malakar, A., Pradhan, M., Minz, A., Banerjee, S., & Sarkar, A.(2024). Tree Improvement Programme for Mahua (Madhuca longifolia) in Eastern India. Biotica Research Today, 6(10), 452-455. https://bioticapublications.com/journal-backend/articlePdf/50129a1ac1.pdf
10. Kumar, J., Nayak, H., & **Sinha, A.** (2024). Evaluation of Seed Germination Parameters of Kalmegh Genotypes in Nursery. International Journal of Bio-resource and Stress Management, 15(Apr, 4), 01-10. https://doi.org/10.23910/1.2024.5240

11. Mishra , Y., **Sinha , A.**, Kumar , R., & Malakar , A. (2024). Effect of Pre-treatments on Germination of *Pterocarpus marsupium* Roxb. for Developing Field Gene-Banks in Eastern India. *Journal of Advances in Biology & Biotechnology*, *27*(4), 151–161. https://doi.org/10.9734/jabb/2024/v27i4748
12. Kumar, J., Nayak, H., & **Sinha, A.** (2024). Screening of suitable Kalmegh genotypes through its seed germination test. *International Journal of Advanced Biochemistry Research, 8*(3), 301-307. https://doi.org/10.33545/26174693.2024.v8.i3d.732

13. **Sinha, A.,** Malakar, A., Banerjee, S., Ehrar, O., & Hembrom, J. P. (2024). Evaluation of teak (Tectona grandis Linn. f) stands for establishing seed production areas in West Bengal. *Environment Conservation Journal*, *25*(2), 443-451. https://doi.org/10.36953/ECJ.25822735
14. Pradhan, M., Malakar, A. & Sinha, A. (2024). Appraisal of the potential habitat distribution of *Madhuca longifolia* manifested remarkable resilience under various socio-climatic scenarios pan-India. *Modelling Earth Systems and Environment*, *10*, 2435–2446. https://doi.org/10.1007/s40808-023-01913-0
15. Kumari, R., Sinha, A., & Mishra, Y. (2023). Importance of conserving *Vitex peduncularis*: A depleting forest resource with immense ethnopharmacological uses. *Pharma Innovation*, *12*(5):119-125.
16. Sarkar, P. K., **Sinha, A.,** Dhakar, M. K., Das, B., & Bhatt, B. P. (2023). Standardization of grafting technique in Kusum [*Schleichera oleosa* (Lour.) Oken]. *Forest Science and Technology*, 1-8. https://doi.org/10.1080/21580103.2023.2166132
17. Malakar, A., Sahoo, H., **Sinha, A.**, & Kumar, A. (2023). Necessity of genetic diversity study and conservation practices in chironji (*Buchanania cochinchinensis* (Lour.) M.R. Almedia). *Environment Conservation Journal*, *24*(1), 253–260. https://doi.org/10.36953/ECJ.12802358
18. Sarkar, P. K., **Sinha, A.,** Kumar, R., Sarkar, A., Banerjee, S., Dhakar, M. K., Das, B. & Bhatt, B. P. (2022). Standardization of Protocols for Plant Growth Media and Seasons for Cuttings of Kusum [*Schleichera oleosa* (Lour.) Oken]. *International Journal of Environment and Climate Change*, *12*(12), 1273-1284. https://doi.org/10.9734/ijecc/2022/v12i121566
19. Warrier, R. R., **Sinha, A.,** Thakur, A., Singh, B., Shirin, F., & Yasodha, R. (2022). Smallholder teak agroforestry in the globalising world: Opportunities and challenges for India. *Agriculture and Forestry Journal*, *6*(1), 32-40.
20. Warrier, R. R., **Sinha, A.,** Thakur, A., Singh, B., Shirin, F., & Yasodha, R. (2022). Small Holder Teak Agrolorsstry Plantations: Scope and Prospects in India. Wood is Good, 3(1), 27-29. Published by Institute of Wood Science and Technology, Bengaluru.
21. Sarkar, P. K., Sinha, A., Das, B., Dhakar, M. K., Shinde, R., Chakrabarti, A., & Bhatt, B. P. (2022). Kusum (Schleichera oleosa (Lour.) Oken): A potential multipurpose tree

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22. Chakraborty, S., Ghosh, P., Sheshshayee, M.S., Viswanath, S., **Sinha, A.** & Shettappanavar, V.S. (2021). Screening of Potential Host of Sandalwood Using Suitable Isotope (15N) Technique under Nursery Conditions. *eJournal of Applies Forest Ecology (eJAFE), 9*(1):9-15.
23. Kumar, R., **Sinha, A.**, Sarkar, P.K. and Kumar, J. 2020. *In vitro* Shoot Proliferation from Nodal Explants of *Aegiceras corniculatum* L. (Blanco.). Int. J. Curr. Microbiol. App. Sci., 9(11):3113-3119.
24. Chakraborty, S., Manjunatha, L., Shetteppanavar, V.S., Devakumar, A.S., Viswanath, S. and **Sinha, A.** 2020. Screening of Selected Hosts for Sandalwood Seedlings at Nursery Stage Based on Host Cation Exchange Capacity. Indian Forester, 146(12): 999-1008.
25. Kumar, R., **Sinha, A.**, Sarkar, P.K. and Kumar, J. 2020. *In vitro* callus induction from nodal and leaf explants of *Heritiera fomes* Buch- Ham: an endangered mangrove. International Journal of Chemical Studies, 8(1): 2785-2788.
26. Sarkar, P.K., **Sinha, A.,** Bhatt, B.P., Kumar, R., Sarkar, A., Banerjee, S., Ghosh, J., Kujur, R.S. and Kulkarni, N. 2019. Effect of pretreatment of explants and *in vitro* culture media in controlling infection and browning in Kusum [*Schleichera oleosa* (Lour.) Oken]. International Journal of Chemical studies, SP6: 914-917.
27. Sarkar, P.K., **Sinha, A.**, Das, B., Shinde, R., Dhakar, M.K. and Das, B. 2019. Bamboo plantation: a step forward in doubling farmer's income in eastern India. Agriculture & Food: e-Newsletter, 1(2): 1-5.
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29. **Sinha, A**., Kumar, J., Nirala, D.P.,Ambasta, N., Kumar, R. and Kumari, P. 2018. Fruit setting and their seed germination rate of half diallel crossed flower of eight best selected *Jatropha curcas*. Journal of Pharmacognosy and Phytochemistry, SP1: 1966-1969.
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32. Nirala, D.P., **Sinha, A**., Kumar, J., Mukherjee, D., Bardani, Kumar, R. and Kumar, M. 2018. Effect of apical pruning and foliar spray of potassium nitrate on seed yield and oil content of *Jatropha curcas* L. in agroclimatic zone of Jharkhand. Journal of Pharmacognosy and Phytochemistry, SP1: 1177-1180.

33. Nayak, H., **Sinha, A**., Bhol, N. and Kumar, J. 2017. Variation in productivity and progeny quality of some teak clones. Progressive Research– An International Journal, 12(Special III): 2160-2164.
34. Nayak, H., **Sinha, A.** Bhol, N. and Kumar, J. 2016. Assessment of variability in planting stock quality of open pollinated seeds of teak clones. J Tree Sci., 35(1): 39-45.
35. Kumar, J., **Sinha, A.**, Kumar, N., Pande, A. and Nayak, H. 2015. Comparative assessment of seed germination parameters of *Andrographis paniculata* (Kalmegh) under laboratory conditions. *Annals of Plant and Soil Research*, **17** (5): 155-160.
36. Kumar, J., **Sinha, A.**, Kumar, N., Pande, A., Mahto, C. S. and Nayak, H. 2015. Evaluation of genetic diversity in Kalmegh (*Andrographis paniculata*) germplasms through growth, yield and reproductive parameters. *Annals of Plant and Soil Research*, **17** (5): 161-166.
37. **Sinha, A.**, Das, R., Deka, B., Viswanath, S., Chandrashekar, B.S. and Chakraborty, S. 2014. Authentication , Micropropagation and Conservation of *Embelia ribes* - a vulnerable medicinal plant. Indian Forester, 140(7): 707-714.
38. **Sinha, A.** 2014. Sundarban Bagh sanrakshan chhetra ke pratirodhak chhetra me mukshya briksh jatiya (in Hindi). *Sodhtaru*, 1(1): 14-20.
39. Lal, H.S., Singh, S. and **Sinha, A.** 2014. Vegetational diversity and its change dynamics in Indian Sundarbans. *Indian journal of Forestry*, 37(4): 365-370.
40. Kumar, R., Kumari, A., **Sinha, A.**, and Maurya, S. 2014. New report of *Lasiodiplodia theobromae* causing *Jatropha* decline in Eastern Platue and Hill region of India. *Archives of Phytopathology and Plant Protection*, **47** (11): 1286-1290.
41. Pandey, A. and **Sinha, A.** 2013. Effects of mannitol, sorbitol and sucrose on growth inhibition and in vitro conservation of germplasm of *Asparagus racemosus*-an important medicinal plant. Medicinal Plants-International Journal of Phytomedicines and Related Industries, 5: 71-74.
42. **Sinha, A.** 2012. Clonal propagation of *Schleichera oleosa* by cleft grafting. Indian Forester, 138 (9): 835-838.
43. Supriya Kumari; Kumar, R.; Chakrovourty, S.K.; Chandra, R.; **Sinha, A.** and Nath, S. 2012. Effect of growth promoting substances and rhizome separation technique on clonal propagation of *Bambusa vulgaris* var. *striata*. Indian Forester, 138(2): 116-122.
44. **Sinha, A**. and Mandal, A. K. 2012. *Schleichera oleosa* – a case for Transboundary conservation. IUFRO World Series, 30: 25-29.
45. Divakara, B.N.; **Sinha, A.** and Alur, A. 2011. Genetic variability and character association among seed traits of *Madhuca latifolia* Macb. Indian Forester, 137(7): 860-867.
46. **Sinha, A.** 2010. Exploring the feasibility of bamboo and vegetable intercropping in Jharkhand, India. APA News Asia-Pacific Agroforestry Newsletter No. 37, December 2010. pp. 5-6.

47. **Sinha, A.** 2009. Production potential of some winter vegetables under edible bamboo (*Dendrocalamus asper*). Journal of Bamboo and Rattan, 8(1&2): 91-94.
48. Nath, S., Das, R., Chandra R. and **Sinha, A.** 2009. Bamboo Based Agroforestry for Marginal Lands with Special Reference to Productivity, Market Trend and Economy. Envis –Jharkhand News, March 2009. pp 80 – 96.
49. **Sinha, A.** 2009. Edible bamboo based agroforestry system – an option of livelihood improvement in Jharkhand. (http://www.jharenvis.nic.in/envi_isuue_asinha.html)
50. **Sinha, A.** and Akhtar, S. 2008. Nodal bud culture in *Schleichera oleosa*: aseptic culture establishment, explant survival and influence of plant growth regulators. Indian J Genet., 68(2):219-221.
51. **Sinha, A.** and Lal, J.P. 2007. Effect of mutagens on M_1 parameters and quantitative changes induced in M_2 generation in lentil. Legume Research, 30 (3): 180 – 185.
52. **Sinha, A.** and Prasad, K.G. 2003. Status and Strategies for teak improvement in North East India. Indian Forester, 129 (9):1132-1140.
53. Ghosh, S.K. and **Sinha, A.** 2001. Transgenic Technology. Kisan World, 28(3): 19-20.
54. **Sinha, A.** and Mishra, S.N. 1997. Combining ability analysis in varietal crosses of maize. Indian J Genet., 57(2):149-153.

Book/Book Chapter/ Technical Bulletin

1. **Sinha, A.** (2025). Forest, Forestry, and State of Forests. In: Mandal, A.K., Nicodemus, A. (eds) *Textbook of Forest Science* (pp. 3-21). Singapore: Springer Nature. https://doi.org/10.1007/978-981-97-8289-5_1
2. **Sinha, A.** (2025). Mahua (*Madhuca longifolia*): Tree Improvement Approaches in Eastern India. ICFRE-IFP, Ranchi.
3. Shirin, F., **Sinha, A.,** et al. (2023). Indigenous Traditional Knowledge (ITK) of *Madhuca longifolia* (Mahua) Prevalent in India. ICFRE-TFRI, Jabalpur.
4. Sinha, A. (2022). Seed Stands in Forests of West Bengal. IFP, Ranchi (Publication No.: IFP/BOOK/01/2022).
5. Cultivation of Selected NWFPs – a Manual. ICFRE, Dehradun (2019) (as a contributor)
6. Kumar, S., Kumar, J., Sharma, J.P., Nedunchezhiyan, M., Sarkar, P.K., Sarkar, P and **Sinha, A.** 2016. Crop Diversification Options. In: A.K. Singh, S.S. Mali, B. Das, P. Bhavana, P.R. Kumar, B. Sarkar and B.P. Bhatt (eds). Approaches to Improve Agricultural Water Productivity. Satish Serial Publishing House, Delhi. 145-173.
7. Kumar, A. Rathore, T.S., Palanisamy, K. and Viswanath, S. (2016). ICFRE State of Knowledge Series – III: Advances in Tree Improvement and Forest Genetic Resources Conservation and Management. ICFRE, Dehradun. (as a contributor)
8. Singh, D., Banerjee, C., **Sinha, A.**, Nirala, D.P., Prasad, S. and Bandopadhyay, R. 2014. Analysis of physical properties and biodiesel production from different accessions of *Jatropha curcas*. In: Recent Advances in Bioenergy Research, Vol. III (Eds. Sachin Kumar, A.K. Sarma, S.K. Tyagi, Y.K. Yadav). *Sardar Swaran Singh National Institute of Renewable Energy, Kapurthala.*
9. **Sinha, A.**, Das, R., Banerjee, S., Chakraborty, S. and Das, D.K. 2013. Tree Improvement and Management of Kusum (Schleichera oleosa (Lour.) Oken) for

higher lac production. In: Prospects of Scientific Lac Cultivation in India (eds Kumar, A. and Das, R.). IFP, Ranchi, pp 59-72.

10. **Sinha, A.**, Das, R., Chandra, A., Chakraborty, S. and Sarkar, A. 2013. Conservation of *Embelia ribes* – a vulnerable medicinal plant. In: Assessment & Conservation of Forest Genetic Resources (eds Singh, S. and Das, R.). IFP, Ranchi, pp 183-190.
11. Forest Types of India Revised. ICFRE, Dehradun (2013) (as a Field Trip member)
12. **Sinha, A.** 2011. Cleft grafting in Kusum (*Schleichera oleosa*) – an important lac host. In: Forestry in the Service of Nation: ICFRE Technologies. (V.K. Bahuguna, R. Kumar, R.P. Singh and R. Mishra eds) ICFRE publication, pp. 252-253.
13. **Sinha, A.** and Akhtar, S. 2008. *In vitro* shoot proliferation from nodal explants of mature trees of Kusum (*Schleichera oleosa*). In: Forest Biotechnology in India (eds Ansari, S.A., Narayanan, C. and Mandal, A.K.). Satish Serial Publishing House, Delhi, pp 205 - 213.
14. **Sinha, A.** 2005. Selection of plus trees of *Schleichera oleosa* (Lour.) Oken - a multipurpose tree. In: Multipurpose Trees in the Tropics: Managenemt and Improvement Strategies (eds Tewari, V.P. and Srivastave, R.L.). AFRI, Jodhpur, pp 719 – 721.

Projects implemented as a Principal Investigator/ Co-P.I.

- Genetic Evaluation of *Schleichera oleosa* CPTs for oil and oil yielding traits — 2021–2025 — ICFRE.
- Technical assistance for setting up of Tissue Culture Lab at Garkhatanga, Ranchi — 2021–2023 — SFD, Jharkhand.
- All India Coordinated Research Project on Bamboo (AICRP-2) — 2020–2025 — CAMPA.
- National Program for Conservation and Development of Forest Genetic Resources — 2020–2025 — CAMPA.
- Genetic Improvement of *Azadirachta indica* A. Juss. (Neem) (AICRP-26) — 2020–2025 — CAMPA.
- Genetic Improvement and Value Addition of *Madhuca longifolia* (AICRP-23) — 2020–2025 — CAMPA.
- Quality Teak Production: Capitalizing on Cloning (AICRP-9) — 2020–2025 — CAMPA.
- Preparation of DPR of Mahanadi river through forestry interventions — 2019–2020 — ICFRE.
- Strengthening of Tissue Culture Laboratory for Mass Multiplication of Bamboos — 2018–2021 — ICFRE-NBM.
- Evaluation of the existing Seed Stands and Seed Production Areas (SPA) of major forest trees under the Forest Directorate, West Bengal — 2018–2021 — Directorate of Forests, Govt. of W.B. under JICA fund.
- Bamboo nursery development / management and production of planting materials on a sustainable basis — 2016–2021 — ICFRE-BTSG.
- Improvement in Seed Yield of *Jatropha curcas* through Breeding and Silvicultural Practices — 2014–2018 — DBT, New Delhi.
- Collection, Conservation and Evaluation of *Melia dubia* Germplasm from North-Bengal, Orissa hills and other parts of India for identification and release of superior clones — 2013–2022 — ICFRE.
- Tree Borne Oil seeds (TBOs) in community lands for Improved Livelihoods of Vulnerable Groups of Jharkhand — 2012–2017 — ICFRE.

- Studies on pollarding and propagation in Kusum (*Schleichera oleosa*) for lac — 2011–2015 — ICFRE.
- Introduction of selected genotypes of Karanj, Kusum and Bamboo as tree components in Agroforestry models in lateritic belt of eastern India — 2011–2015 — ICFRE.
- Multi-locational trail of *Jatropha curcas* in different agro-climatic zones and study of agronomic practices — 2010–2014 — DBT, New Delhi.
- Protocol optimization for in vitro propagation and conservation of *Embelia ribes* Burm F. a vulnerable medicinal plant — 2010–2013 — ICFRE.
- Studies on variability in rooting proficiency in selected genotypes of *Pongamia pinnata* (L.) Pierre — 2010–2012 — ICFRE.
- Improvement of clonal propagation techniques of bamboos and enhancement in field survival — 2007–2011 — ICFRE.
- Studies on reproductive biology and propagation techniques of *Schleichera oleosa* Lour (Oken) an oil bearing trees and important lac host plants — 2006–2010 — ICFRE.
- *Schleichera oleosa* (Lour.) Oken, a lac host: *In vitro* Propagation — 2003–2006 — DBT, New Delhi.
- Multilocation field trial of tissue culture raised plantlets of *Dendrocalamus asper* — 2002–2006 — ICFRE.